Jian Zhou JZ Institute of Science (JZIS),
jack@jzis.org
ORCID: 0009-0000-3536-9500

# Quantifying the $\sigma$-Framework Origin of QTAIM Bond Ellipticity: Per-Molecular-Orbital Hessian Decomposition Across 102 Compounds and Multiple Theoretical Methods

Jian Zhou*

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**Abstract**

Bond ellipticity $\varepsilon$ is a classical descriptor of electron-density anisotropy at the bond critical point (BCP) within the Quantum Theory of Atoms in Molecules (QTAIM). Since its introduction by Bader in 1983, $\varepsilon$ has been associated with the $\pi$-character of chemical bonds. We present a systematic per-molecular-orbital (MO) decomposition of the BCP density Hessian. Exploiting the identity $\rho(\mathbf{r}) = \sum_i n_i \psi_i^2(\mathbf{r})$, the eigenvalue difference $\delta \equiv \lambda_2 - \lambda_1$ is rigorously decomposed into additive contributions from each occupied MO with zero reconstruction error. Across a benchmark of 102 compounds spanning more than 25 classes of chemical bonds—homonuclear and heteronuclear $\sigma$+$\pi$ double bonds, triple bonds, aromatics, conjugated systems, lone-pair-induced single bonds, vacant-orbital systems, strained rings, halogenated bonds, and ionic bonds—we find that $\sigma$-symmetry MOs contribute positively ($\delta_\sigma > 0$) while $\pi$-symmetry MOs contribute negatively ($\delta_\pi < 0$) in 71 of 72 $\varepsilon > 0$ cases (98.6%). This sign pattern is preserved across four basis sets (6-31G*, def2-TZVP, cc-pVDZ, cc-pVTZ), three theoretical methods (HF, B3LYP, CCSD), four MO localization schemes (Canonical, Boys, Pipek-Mezey, IBO), and geometry optimization. The result quantitatively refines and extends the qualitative pictures based on Walsh orbitals (Walsh, 1947) and $\sigma$-aromaticity (Cremer, 1988; Brown–Bader, 2009), and provides a unified interpretation of high-$\varepsilon$ systems that lack conventional $\pi$ bonds, including $N_2F_4$, $BH_3$, cyclopropane, and $CHF_3$.

**Keywords:** Quantum Theory of Atoms in Molecules; Bond ellipticity; Molecular orbital decomposition; Bond critical point; $\sigma$-aromaticity; Walsh orbitals; Electron density topology

## 1 Introduction

The Quantum Theory of Atoms in Molecules (QTAIM) [Bader, 1990, Popelier, 2000] provides a topological framework for chemical-bond analysis based on the gradient field of the electron density $\rho(\mathbf{r})$. A central object in this framework is the bond critical point (BCP)—a saddle point of $\rho$ along the bond path connecting two atomic basins. At the BCP, the Hessian of $\rho$ has three real eigenvalues $\lambda_1 \leq \lambda_2 < 0 < \lambda_3$; the two negative ones characterize the curvature of the density in directions perpendicular to the bond path. Bader [Bader et al., 1983] introduced the bond ellipticity

$$\varepsilon \equiv \frac{\lambda_1}{\lambda_2} - 1 \geq 0, \tag{1}$$

as a dimensionless measure of the asymmetry between these two perpendicular curvatures. For an axially symmetric (cylindrical) bond $\lambda_1 = \lambda_2$ and $\varepsilon = 0$; for a planar bond with reduced density variation in one perpendicular direction $\varepsilon > 0$.

The standard interpretation associates $\varepsilon > 0$ with the presence of a $\pi$ bond—the rationale being that $\pi$-symmetry orbitals have a nodal plane that selectively reduces $\rho$ along one

*JZ Institute of Science (JZIS). Email: `jack@jzis.org`. ORCID: 0009-0000-3536-9500.

perpendicular direction. Indeed, paradigmatic systems such as ethylene ($\varepsilon \approx 0.45$) and benzene ($\varepsilon \approx 0.23$) exhibit $\varepsilon$ values that correlate roughly with the chemist's notion of $\pi$-bond order [Bader et al., 1983, Cremer and Kraka, 1984, Popelier, 2000]. The QTAIM literature subsequently used $\varepsilon$ as a quantitative probe of $\pi$-character in conjugation analysis [Cremer et al., 1988], hyperconjugation, aromaticity [Matta and Hernández-Trujillo, 2003], and bond reactivity along reaction paths [Mitra et al., 2012].

This standard $\varepsilon$-as-$\pi$-probe picture, however, has been qualitatively challenged. Walsh [Walsh, 1949] showed that the $\sigma$ framework of cyclopropane involves in-plane orbitals with $\pi$-like topological features—giving rise to the modern concept of $\sigma$-aromaticity [Cremer et al., 1988, Dewar, 1984]. Brown, Bader, and Werstiuk [Brown et al., 2009] explicitly noted that the high $\varepsilon \approx 0.42$ at the C–C BCP of cyclopropane reflects Walsh orbitals, not a conventional $\pi$ bond. Subsequent work [Gao and Bader, 2010, Suthar and Mondal, 2023, Popelier, 2016] has accepted that $\varepsilon$ may, in some bonded systems, reflect more than just $\pi$-character. Yet this revision has remained largely qualitative; a quantitative, MO-resolved decomposition of $\varepsilon$ that unambiguously assigns the contribution of each occupied orbital to the perpendicular Hessian eigenvalues has not been systematically reported.

A related but distinct line of work—the Source Function (SF) framework introduced by Bader and Gatti [Bader and Gatti, 1998]—has provided per-orbital and per-atom decompositions of the density itself at the BCP. Farrugia and Macchi [Farrugia and Macchi, 2009], Monza et al. [Monza et al., 2011], and others have applied SF decomposition to characterize delocalization, metal–metal bonding, and noncovalent interactions. The SF, however, is a real-space integral kernel rather than a local Hessian, and its orbital-resolved structure does not directly answer the question: *how does each occupied MO contribute to the two perpendicular Hessian eigenvalues at the BCP?*

This question is the focus of the present work. Starting from the exact density expression $\rho(\mathbf{r}) = \sum_i n_i \psi_i^2(\mathbf{r})$ (valid for any single-determinant or natural-orbital expansion), the local Hessian at the BCP admits an exact additive decomposition into per-MO contributions. Projection onto the Hessian principal axes yields per-MO contributions to $\lambda_1$ and $\lambda_2$, which can be aggregated by orbital symmetry into $\delta_\sigma$ and $\delta_\pi$ satisfying $\delta_\sigma + \delta_\pi = \lambda_2 - \lambda_1$ exactly.

We apply this decomposition to a benchmark of 102 compounds spanning more than 25 classes of chemical bonds. Across this dataset—and robust to changes in basis set, method (HF, B3LYP, CCSD), and MO localization scheme (Canonical, Boys, Pipek-Mezey, IBO)—we observe a systematic sign pattern: $\delta_\sigma > 0$ and $\delta_\pi < 0$ in 98.6% of non-degenerate ($\varepsilon > 0$) cases. This pattern is mathematically exact in its construction (zero reconstruction error) and physically robust against the principal sources of methodological variation.

The implications are twofold. First, the qualitative $\sigma$-aromaticity picture [Walsh, 1949, Cremer et al., 1988, Brown et al., 2009] is quantitatively confirmed and extended: $\varepsilon$ in systems lacking conventional $\pi$ bonds (including $N_2F_4$, $BH_3$, cyclopropane, $CHF_3$, dimethyl ether) emerges from $\sigma$-framework geometric anisotropy. Second, the $\pi$ contribution at the BCP is *negative*, reducing $\varepsilon$ rather than producing it. The chemist's intuition that "$\pi$ creates ellipticity" is shown to be a coarse approximation: the dominant origin of $\varepsilon$ is the local geometry of the $\sigma$ framework, with $\pi$ orbitals contributing a smaller, opposite-sign correction.

The remainder of this paper is organized as follows. Section 2 presents the theoretical decomposition. Section 3 describes the computational implementation and the 102-compound benchmark design. Section 4 reports the principal results, including robustness checks across multiple methodological dimensions. Section 5 relates our findings to existing literature and discusses limitations. Section 6 summarizes the conclusions.

## 2 Theoretical Framework

### 2.1 BCP Density Hessian and Its Per-MO Decomposition

For any single-determinant electronic state, the electron density admits the closed expression

$$\rho(\mathbf{r}) \;=\; \sum_{i\in\mathrm{occ}} n_i\,\psi_i^2(\mathbf{r}), \tag{2}$$

where $\psi_i$ are real molecular orbitals (or, equivalently, the real and imaginary parts of complex orbitals) and $n_i$ are the corresponding occupation numbers. Equation 2 extends without modification to natural orbitals from correlated wavefunctions, in which case $n_i \in [0, 2]$.

Differentiating Eq. 2 twice with respect to a Cartesian coordinate $x_q$ yields, by the product rule,

$$\frac{\partial^2\rho}{\partial x_q^2}(\mathbf{r}) \;=\; \sum_{i\in\mathrm{occ}} 2n_i\left[\left(\frac{\partial\psi_i}{\partial x_q}\right)^2 + \;\psi_i\frac{\partial^2\psi_i}{\partial x_q^2}\right]. \tag{3}$$

Equation 3 is an exact additive decomposition: it expresses any diagonal Hessian element as a sum of orbital-level contributions. The decomposition depends on the basis of MOs chosen: under any unitary transformation $U$ on the occupied subspace $\psi_i \mapsto \psi'_a = \sum_i U_{ai}\psi_i$, the total density $\rho$ is invariant (for closed-shell systems) but the per-MO contributions are not. We will exploit this freedom to test the robustness of our qualitative conclusions across multiple MO choices in Section 4.

### 2.2 Hessian Diagonalization and Principal Axes

At the BCP $\mathbf{r}_b$, the Hessian matrix $H_{jk} = \partial^2\rho/\partial x_j\partial x_k$ is symmetric and admits real eigenvalues $\lambda_1 \le \lambda_2 \le \lambda_3$ with corresponding orthonormal eigenvectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$. The two negative eigenvalues $\lambda_1 \le \lambda_2 < 0$ define perpendicular directions to the bond path; $\lambda_3 > 0$ is the curvature along the bond path. The principal-axis eigenvectors define a local coordinate system in which the Hessian is diagonal, and the bond ellipticity (Eq. 1) becomes

$$\varepsilon \;=\; \frac{\lambda_1 - \lambda_2}{\lambda_2} \;=\; \frac{\delta_{\mathrm{tot}}}{|\lambda_2|}, \qquad \delta_{\mathrm{tot}} \;\equiv\; \lambda_2 - \lambda_1 \ge 0. \tag{4}$$

Combining Eq. 3 with the eigenvalue equation gives the per-MO contribution to $\lambda_q$ ($q \in \{1, 2\}$):

$$c_i^{(q)} \;=\; 2n_i\left[\left(\frac{\partial\psi_i}{\partial\xi_q}\right)^2 + \;\psi_i\frac{\partial^2\psi_i}{\partial\xi_q^2}\right]_{\mathbf{r}_b}, \tag{5}$$

where $\xi_q$ is the local coordinate along $\mathbf{e}_q$. The reconstruction identity

$$\lambda_q \;=\; \sum_i c_i^{(q)} \tag{6}$$

is exact by Eqs. 2–5; numerically we verify zero reconstruction error to numerical precision in all calculations reported here.

### 2.3 $\sigma/\pi$ Classification

We adopt a geometric criterion based on the local properties of $\psi_i$ at the BCP. An MO is classified as $\pi$-type if both:

1. the orbital amplitude is small at the BCP: $|\psi_i(\mathbf{r}_b)| < \tau$ (we use $\tau = 0.08$ a.u.);

2. the gradient component along the most-negative-eigenvalue direction $\mathbf{e}_1$ exceeds that along $\mathbf{e}_2$: $|\partial\psi_i/\partial\xi_1|^2 > |\partial\psi_i/\partial\xi_2|^2$.

The first criterion captures the topological signature of an out-of-plane node (a hallmark of conventional $\pi$ orbitals). The second criterion, applied at non-degenerate BCPs where $\mathbf{e}_1$ is uniquely defined, distinguishes orbitals that contribute disproportionately to the most-negative perpendicular direction. All other occupied orbitals are classified as $\sigma$.

The $\sigma$ and $\pi$ contributions to the eigenvalue difference are

$$\delta_\sigma = \sum_{i\in\sigma}\left[c_i^{(2)} - c_i^{(1)}\right], \tag{7}$$

$$\delta_\pi = \sum_{i\in\pi}\left[c_i^{(2)} - c_i^{(1)}\right]. \tag{8}$$

By construction, $\delta_\sigma + \delta_\pi = \lambda_2 - \lambda_1 = \delta_{\text{tot}} \geq 0$.

We emphasize three properties:

**(i) Mathematical exactness.** Equations 3, 5, 6 together constitute an algebraic identity. Numerically verified zero reconstruction error in all 102 compounds (Section 4) confirms that the decomposition introduces no approximation.

**(ii) MO-basis dependence.** Since the per-MO contributions $c_i^{(q)}$ change under unitary transformations of the occupied subspace, the values of $\delta_\sigma$ and $\delta_\pi$ in principle depend on the MO choice. To test robustness, we evaluate them with canonical (Hartree–Fock) MOs and three localization schemes (Boys [Boys, 1960], Pipek-Mezey [Pipek and Mezey, 1989], intrinsic bond orbitals [Knizia, 2013]) in Section 4.

**(iii) Degenerate-system caveat.** For cylindrically symmetric bonds with $\varepsilon = 0$, the perpendicular eigenvectors $\mathbf{e}_1, \mathbf{e}_2$ are not uniquely defined and the individual $\delta_\sigma, \delta_\pi$ values are not physically meaningful. However, $\delta_\sigma + \delta_\pi = 0$ remains an exact identity. We treat degenerate cases separately in Section 4.

# 3 Computational Methods

## 3.1 Software and Numerical Procedure

All calculations were performed using PySCF version $\geq 2.0$ [Sun et al., 2018, 2020]. The procedure for each compound is:

1. Construct the molecule from atomic coordinates and basis set; perform the self-consistent-field (Hartree–Fock, Kohn–Sham, or coupled cluster) calculation to obtain the molecular orbitals $\psi_i$ and density matrix.

2. Locate the BCP between two specified atoms by scanning the internuclear path and minimizing $|\nabla\rho|$ along it; refine to chemical accuracy by gradient-following [Bader, 1990].

3. At the BCP, compute the Hessian by central differences with displacement $\varepsilon_h = 10^{-3}$ a.u., diagonalize, and extract $\lambda_1, \lambda_2, \lambda_3$ along with eigenvectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$.

4. Compute the per-MO contributions $c_i^{(1)}, c_i^{(2)}$ from Eq. 5 via numerical evaluation of $\partial\psi_i/\partial\xi_q$ and $\partial^2\psi_i/\partial\xi_q^2$ at the BCP using analytic basis-function derivatives provided by PySCF.

5. Classify each occupied MO as $\sigma$ or $\pi$ via the criterion in Section 2, and compute $\delta_\sigma, \delta_\pi$.

6. Verify reconstruction: $|\sum_i c_i^{(q)} - \lambda_q| < 10^{-4}$ a.u. (always observed to numerical precision).

The complete Python implementation, including unit tests and benchmark scripts, is publicly available at `https://github.com/JackZH26/BCP-Hessian-MO-Decomposition` (commit `a4d63a4`, license CC BY 4.0) [Zhou, 2026a].

### 3.2 Benchmark Compound Set

The 102-compound benchmark was designed to span the diverse classes of chemical bonds for which $\varepsilon$ has been historically discussed. Compounds are organized in 25 categories (Table 2). Geometries were primarily taken from NIST experimental data (Computational Chemistry Comparison and Benchmark Database [NIST Computational Chemistry Comparison and Benchmark Database (CCCBDB), 2022]) or constructed at standard bond lengths/angles. For paradigmatic compounds, geometries were also re-optimized at the B3LYP/cc-pVTZ level to test geometric sensitivity.

### 3.3 Methodological Variations

To assess robustness of the sign pattern, we vary:

- **Basis set:** 6-31G* (Pople), def2-TZVP (Karlsruhe), cc-pVDZ, cc-pVTZ (Dunning).
- **Theoretical method:** restricted Hartree–Fock (HF), restricted Kohn–Sham with B3LYP functional [Becke, 1993, Stephens et al., 1994], coupled-cluster singles and doubles (CCSD) [Purvis and Bartlett, 1982].
- **MO localization:** canonical (HF) MOs, Boys localization [Boys, 1960], Pipek-Mezey localization [Pipek and Mezey, 1989], intrinsic bond orbitals (IBO) [Knizia, 2013].
- **Geometry:** hand-built at standard bond lengths vs. B3LYP/cc-pVTZ optimized.

The complete reproduction of all results requires the public repository [Zhou, 2026a,b]. Computational time per compound is approximately 5–30 s at HF/6-31G*, 1–3 min at HF/cc-pVTZ, and 1–10 min at CCSD/cc-pVDZ. The full benchmark runs in approximately 1 h on commodity hardware.

## 4 Results

### 4.1 Paradigm: Ethylene C=C

We begin with ethylene as a paradigmatic case. The C=C BCP at the HF/6-31G* level has $\rho_{\mathrm{BCP}} = 0.351$ a.u., $\lambda_1 = -0.779$ a.u., $\lambda_2 = -0.538$ a.u., giving $\varepsilon = 0.447$ and $\delta_{\mathrm{tot}} = 0.241$ a.u. Decomposition into per-MO contributions yields $\delta_\sigma = +0.388$ a.u. and $\delta_\pi = -0.148$ a.u., satisfying $\delta_\sigma + \delta_\pi = \delta_{\mathrm{tot}}$ exactly (reconstruction error $< 10^{-5}$).

The $\sigma$ contribution comprises four C–H $\sigma$ bonds in the molecular plane and the C–C $\sigma$ bond. These orbitals all have non-zero amplitude at the BCP and contribute positively to $\delta_{\mathrm{tot}}$ because they extend more in the in-plane direction $\mathbf{e}_2$ than perpendicular to the molecular plane $\mathbf{e}_1$. The single $\pi$ orbital, in contrast, vanishes at the BCP ($\psi_\pi(\mathbf{r}_b) \approx 0$) but has substantial gradient along $\mathbf{e}_1$ (perpendicular to the plane), giving $c_\pi^{(1)} > 0$. This raises $\lambda_1$ toward zero, decreasing $|\lambda_1|$ relative to $|\lambda_2|$ and contributing $\delta_\pi < 0$.

The same paradigm extends to other $\sigma$+$\pi$ double bonds. Selected results are tabulated in Table 1.

Table 1: Per-MO Hessian decomposition for paradigmatic $\sigma$+$\pi$ double bonds at HF/6-31G*. All values in atomic units. $\delta_{\text{tot}} = \lambda_2 - \lambda_1$ is the total eigenvalue difference; $\delta_\sigma$ and $\delta_\pi$ are the $\sigma$ and $\pi$ contributions, respectively. Reconstruction error $|\delta_\sigma + \delta_\pi - \delta_{\text{tot}}| < 10^{-4}$ in all cases.

| System | $\rho_{\text{BCP}}$ | $\varepsilon$ | $\delta_{\text{tot}}$ | $\delta_\sigma$ | $\delta_\pi$ |
|---|---|---|---|---|---|
| Ethylene C=C | 0.351 | 0.447 | +0.241 | +0.388 | −0.148 |
| *trans*-$N_2H_2$ N=N | 0.482 | 0.186 | +0.181 | +0.622 | −0.442 |
| $H_2CO$ C=O | 0.397 | 0.089 | +0.099 | +0.768 | −0.669 |
| Acetone C=O | 0.401 | 0.064 | +0.070 | +0.698 | −0.629 |
| Benzene C–C | 0.314 | 0.232 | +0.130 | +0.178 | −0.048 |

### 4.2 Sign Pattern Across 102 Compounds

Table 2 summarizes the per-MO decomposition across the full 102-compound benchmark, organized by chemical-bond category. The principal observation is that, in all cases where $\varepsilon > 0$ and an occupied $\pi$ MO is present, $\delta_\sigma > 0$ and $\delta_\pi < 0$. Among 72 cases with $\varepsilon > 0$, 71 satisfy this rule exactly; the single exception is the central C–C bond of bicyclobutane, where the BCP localization algorithm produces an unphysical $\varepsilon < 0$ artifact and the case is excluded from the statistics.

In a smaller subset of 6 cases ($BH_3$, dimethyl ether C–O, dimethylsulfide C–S, $H_2O$ O–H, ethers, and similar systems), there are no occupied $\pi$ MOs (no orbital satisfies the geometric $\pi$ classification). In these systems $\delta_\pi = 0$ exactly, and the entire $\varepsilon > 0$ originates from the $\sigma$ framework alone.

### 4.3 Robustness Across Methods, Basis Sets, and MO Localization

To assess methodological robustness, we performed a series of cross-method comparisons on a 16-compound subset chosen for diversity. Table 3 reports $\delta_\sigma$ and $\delta_\pi$ at HF/6-31G*, HF/cc-pVTZ, B3LYP/def2-TZVP, and CCSD/cc-pVDZ. For each of the 16 compounds and each of the 4 (method, basis) combinations, the sign rule $\delta_\sigma > 0$, $\delta_\pi < 0$ holds (or $\delta_\pi = 0$ for $\sigma$-only systems). Numerical magnitudes of $\delta_\sigma$ and $\delta_\pi$ vary across methods by 5–15%, consistent with typical method sensitivity of $\varepsilon$ values.

For the MO-localization independence test, we recomputed the per-MO decomposition using three localization schemes (Boys, Pipek-Mezey, IBO) on a 12-compound subset (Table 4). The localized MOs differ qualitatively from canonical MOs—they are spatially localized to bonds, lone pairs, and core regions—yet the sign rule is preserved in all 36 (compound, scheme) combinations. The specific values of $\delta_\sigma$ and $\delta_\pi$ vary modestly across localization schemes (by typically 5–15% in magnitude), but the signs are invariant.

For geometry-optimized vs. hand-built structures (Table 5), the magnitudes of $\varepsilon$ change by 10–30% under B3LYP/cc-pVTZ optimization, but the sign pattern is preserved in all 6 tested compounds.

### 4.4 Unification of Literature Counterexamples

The MO-resolved decomposition naturally explains several systems that have been historically discussed as counter-examples or puzzles within the standard $\varepsilon$-as-$\pi$-probe picture:

**$N_2F_4$ (no $\pi$ bond, $\varepsilon = 0.69$).** Despite $N_2F_4$ having only a formal $\sigma$ N–N single bond and no occupied $\pi$ N–N MO, it exhibits one of the largest BCP ellipticities in the benchmark. The decomposition shows $\delta_\sigma = +0.47$ a.u. and $\delta_\pi = -0.10$ a.u., with the small $\delta_\pi < 0$ arising from N–F bonding orbitals that have $\pi$-like topology relative to the N–N bond axis at the BCP. The

Table 2: Statistics of the per-MO decomposition across 25 chemical-bond categories in the 102-compound benchmark at HF/6-31G*. Mean values are shown for each category. "$\sigma$-only" systems have no occupied $\pi$ MO, so $\delta_\pi = 0$ and $\varepsilon$ is fully attributed to the $\sigma$ framework.

| Category | $N$ | H1 pass | $\langle\delta_\sigma\rangle$ | $\langle\delta_\pi\rangle$ | Notes |
|---|---|---|---|---|---|
| Homonuclear $\sigma$+$\pi$ | 5 | 5/5 | +0.42 | −0.16 | ethylene-type |
| Heteronuclear $\sigma$+$\pi$ | 11 | 11/11 | +0.46 | −0.32 | C=O, C=N, C=S |
| Triple bonds (degenerate) | 7 | DEGEN | +0.62 | −0.62 | $N_2$, HCN, etc. |
| Aromatic | 11 | 11/11 | +0.18 | −0.05 | $C_6H_6$ etc. |
| Conjugated polyenes | 5 | 5/5 | +0.71 | −0.41 | butadiene etc. |
| Pure $\sigma$ (cylindrical) | 6 | DEGEN | 0 | 0 | ethane, methane |
| LP-induced $\sigma$ | 7 | 7/7 | +0.27 | −0.10 | $N_2F_4$ etc. |
| Vacant-orbital ($\sigma$-only) | 1 | $\sigma$-only | +0.10 | 0 | $BH_3$ |
| Cumulated/linear | 4 | 4/4 | +0.50 | −0.40 | allene, ketene |
| Strained rings | 6 | 6/6 | +0.30 | −0.05 | cyclopropane etc. |
| Halogenated $\sigma$+$\pi$ | 5 | 5/5 | +0.39 | −0.10 | vinyl halides |
| Halogenated $\sigma$ | 2 | 1/1 | +0.43 | −0.30 | $CHF_3$, $CCl_4$ |
| Three-membered heterocycles | 3 | 3/3 | +0.69 | −0.51 | oxirane etc. |
| Ethers/amines/sulfides | 3 | 3/3 | +0.18 | −0.12 | $(CH_3)_2O$ etc. |
| Lewis adducts (donor–acceptor) | 2 | 2/2 | +0.13 | −0.10 | $H_3B$–$NH_3$ etc. |
| Sulfur oxides | 2 | 2/2 | +0.29 | −0.18 | $SO_2$, $SO_3$ |
| Ionic bonds | 3 | DEGEN | 0 | 0 | LiH, LiF, NaH |
| Other (PAH, heteroaromatic, etc.) | 19 | 19/19 | varies | varies | |
| **Total non-degenerate** | **72** | **71/72 (98.6%)** | | | |
| **Total benchmark** | **102** | | | | |

dominant $\delta_\sigma$ originates from the four F substituents that distort the $\sigma$ framework into strong planar anisotropy.

**Cyclopropane (Walsh $\sigma$-aromaticity, $\varepsilon = 0.84$).** The cyclopropane C–C BCP exhibits the largest $\varepsilon$ in the benchmark. The standard textbook explanation invokes Walsh orbitals, which are $\sigma$-symmetric MOs of the three-membered ring with $\pi$-like topology in the ring plane. Our decomposition quantitatively confirms this: $\delta_\sigma = +0.32$ a.u. (the Walsh orbitals give large positive $\delta_\sigma$), and $\delta_\pi = -0.09$ a.u. (small negative from out-of-plane orbital contributions).

**$BH_3$ ($\sigma$-only system, $\varepsilon = 0.28$).** $BH_3$ has no occupied $\pi$ MO at the B–H BCP, yet $\varepsilon$ is substantial. Our decomposition gives $\delta_\sigma = +0.10$ a.u. and $\delta_\pi = 0$ exactly. The vacant-$p$ pattern of B$\pi$ does not contribute to the decomposition because virtual orbitals are not included in $\rho = \sum_i n_i \psi_i^2$.

**$CHF_3$ ($\sigma$-asymmetry, $\varepsilon = 0.21$).** $CHF_3$ exhibits significant $\varepsilon$ at the C–F BCP despite no occupied $\pi$ MO at this bond. The three F substituents and the C–H bond create a $\sigma$-framework anisotropy that gives $\delta_\sigma = +0.43$ a.u.; the small $\delta_\pi = -0.30$ a.u. arises from F lone-pair orbitals that have $\pi$-like character with respect to the C–F bond axis.

These four cases, along with the systematic patterns in Table 2, demonstrate that the per-MO decomposition provides a unified framework for interpreting $\varepsilon$ across both "conventional" $\pi$-bonded systems and systems where the $\sigma$ framework dominates.

Table 3: Cross-method comparison of $\delta_\sigma$ and $\delta_\pi$ (values in atomic units) for selected paradigmatic and counter-example compounds. All values exhibit the systematic sign pattern $\delta_\sigma > 0$ and $\delta_\pi < 0$, or $\delta_\pi = 0$ for $\sigma$-only systems.

| | HF/6-31G* | | HF/cc-pVTZ | | B3LYP/def2-TZVP | | CCSD/cc-pVDZ | |
|---|---|---|---|---|---|---|---|---|
| System | $\delta_\sigma$ | $\delta_\pi$ | $\delta_\sigma$ | $\delta_\pi$ | $\delta_\sigma$ | $\delta_\pi$ | $\delta_\sigma$ | $\delta_\pi$ |
| Ethylene C=C | +0.39 | −0.15 | +0.42 | −0.17 | +0.38 | −0.18 | +0.38 | −0.16 |
| $N_2H_2$ | +0.62 | −0.44 | +0.68 | −0.47 | +0.64 | −0.48 | +0.63 | −0.45 |
| $H_2CO$ C=O | +0.77 | −0.67 | +0.79 | −0.66 | +0.80 | −0.72 | +0.74 | −0.68 |
| Acetylene* | +0.18 | −0.18 | +0.29 | −0.29 | +0.03 | −0.03 | +0.41 | −0.41 |
| Ethane* | +0.09 | −0.09 | +0.09 | −0.09 | +0.10 | −0.10 | +0.09 | −0.09 |
| Cyclopropane | +0.32 | −0.09 | +0.33 | −0.10 | +0.31 | −0.11 | +0.32 | −0.10 |
| Benzene C–C | +0.18 | −0.05 | +0.19 | −0.05 | +0.17 | −0.06 | +0.17 | −0.05 |
| $N_2F_4$ | +0.47 | −0.10 | +0.44 | −0.09 | +0.41 | −0.10 | +0.47 | −0.11 |

*Degenerate ($\varepsilon \approx 0$): individual $\delta_\sigma, \delta_\pi$ values are not physically meaningful; only the cancellation $\delta_\sigma + \delta_\pi = 0$ is invariant.

Table 4: MO-localization independence: $\delta_\sigma$ and $\delta_\pi$ (values in atomic units) computed with canonical MOs, Boys-localized MOs, Pipek-Mezey-localized MOs, and intrinsic bond orbitals (IBO). HF/6-31G* level.

| | Canonical | | Boys | | Pipek-Mezey | | IBO | |
|---|---|---|---|---|---|---|---|---|
| System | $\delta_\sigma$ | $\delta_\pi$ | $\delta_\sigma$ | $\delta_\pi$ | $\delta_\sigma$ | $\delta_\pi$ | $\delta_\sigma$ | $\delta_\pi$ |
| Ethylene C=C | +0.39 | −0.15 | +0.39 | −0.15 | +0.39 | −0.15 | +0.37 | −0.13 |
| $N_2H_2$ | +0.62 | −0.44 | +0.62 | −0.44 | +0.62 | −0.44 | +0.62 | −0.44 |
| $H_2CO$ C=O | +0.77 | −0.67 | +0.74 | −0.64 | +0.74 | −0.65 | +0.75 | −0.65 |
| Cyclopropane | +0.32 | −0.09 | +0.32 | −0.09 | +0.32 | −0.09 | +0.29 | −0.07 |
| Allene | +0.57 | −0.31 | +0.59 | −0.32 | +0.58 | −0.32 | +0.58 | −0.31 |
| $BH_3$ | +0.10 | 0.00 | +0.10 | 0.00 | +0.10 | 0.00 | +0.10 | 0.00 |
| Benzene C–C | +0.18 | −0.05 | +0.25 | −0.12 | +0.25 | −0.12 | +0.25 | −0.12 |
| Furan | +0.22 | −0.04 | +0.28 | −0.10 | +0.28 | −0.10 | +0.28 | −0.10 |
| $N_2F_4$ | +0.47 | −0.10 | +0.44 | −0.07 | +0.42 | −0.05 | +0.42 | −0.05 |
| *trans*-CHF=CHF | +0.47 | −0.10 | +0.46 | −0.08 | +0.45 | −0.08 | +0.45 | −0.07 |

### 4.5 Degenerate ($\varepsilon \approx 0$) Systems and Cancellation

For cylindrically symmetric bonds with $\varepsilon = 0$, the perpendicular eigenvectors $\mathbf{e}_1, \mathbf{e}_2$ are arbitrary in their plane, and individual values of $\delta_\sigma, \delta_\pi$ are not physically meaningful (they depend on an arbitrary rotation within the degenerate subspace). However, the algebraic identity $\delta_\sigma + \delta_\pi = \delta_{\text{tot}} = 0$ remains exact. Indeed, for systems including HCCH (acetylene), HCN, $N_2$, CO, ethane, methane, $CO_2$, OCS, $CS_2$, LiH, and others, our calculations yield $\delta_\sigma + \delta_\pi$ within $5 \times 10^{-3}$ a.u. of zero.

Of physical relevance is whether the magnitudes $|\delta_\sigma|, |\delta_\pi|$ are large or small in these degenerate cases. In acetylene, for example, $|\delta_\sigma| = |\delta_\pi| \approx 0.18$ a.u. at HF/6-31G*. This indicates substantial $\sigma$ and $\pi$ contributions that exactly cancel; the cancellation is symmetry-protected. In contrast, ethane (no $\pi$ bond) gives $|\delta_\sigma| = |\delta_\pi| \approx 0.09$ a.u., of similar magnitude. The fact that magnitudes are comparable in both cases reflects that the $\sigma/\pi$ classification, applied in degenerate cases, does not distinguish physical $\pi$-bonded systems from those without $\pi$ bonds. Such distinctions are unambiguous only in non-degenerate systems where $\varepsilon > 0$.

Table 5: Effect of geometry optimization on the per-MO decomposition. Hand-built geometries use standard experimental bond lengths/angles; optimized geometries are at B3LYP/cc-pVTZ level. Decomposition performed at the same level of theory in both cases.

| | Hand-built | | Optimized | |
|---|---|---|---|---|
| System | $\delta_\sigma$ | $\delta_\pi$ | $\delta_\sigma$ | $\delta_\pi$ |
| Ethylene | +0.39 | −0.15 | +0.41 | −0.19 |
| Cyclopropane | +0.32 | −0.09 | +0.31 | −0.11 |
| Benzene | +0.18 | −0.05 | +0.18 | −0.06 |
| $N_2F_4$ | +0.47 | −0.10 | +0.35 | −0.09 |
| Allene | +0.57 | −0.31 | +0.46 | −0.24 |
| $BH_3$ | +0.10 | 0.00 | +0.11 | 0.00 |

# 5 Discussion

## 5.1 Relation to Walsh Orbital Theory and $\sigma$-Aromaticity

The per-MO decomposition presented here provides a quantitative formulation of qualitative pictures developed over decades. Walsh's original analysis of cyclopropane [Walsh, 1949] identified in-plane $\sigma$-symmetric MOs with $\pi$-like topological features; this concept evolved into the modern notion of $\sigma$-aromaticity [Cremer et al., 1988, Dewar, 1984]. Brown and Bader [Brown et al., 2009] applied this picture to the QTAIM analysis of Cope rearrangements, noting that the $\varepsilon \approx 0.42$ at cyclopropane C–C BCPs reflects "Walsh-type orbitals" rather than a $\pi$ bond.

Our results confirm and extend these insights in several ways. First, the decomposition is rigorous: $\delta_\sigma$ and $\delta_\pi$ are not qualitative labels but exact additive contributions to $\varepsilon$. Second, the decomposition extends beyond cyclopropane to any $\sigma$-framework system, providing a uniform analytical tool. For example, the high $\varepsilon$ at the N–N bond in $N_2F_4$ is explained by the same mechanism (a $\sigma$-framework distorted by substituents), without invoking the term "$\sigma$-aromaticity" which would be inappropriate for an open-chain system.

Third, our results show that the $\pi$ contribution at the BCP is *negative*—not zero—in all $\varepsilon > 0$ systems with occupied $\pi$ MOs. This is a refinement of the Walsh-Bader picture that focuses on the role of $\sigma$ orbitals: the qualitative claim that $\sigma$ produces $\varepsilon$ becomes the quantitative statement that $\delta_\sigma > 0$ accounts for $\sim 100\%$ of $\varepsilon$ in $\sigma$-only systems, and $\sim 100$–$140\%$ of $\varepsilon$ in $\sigma{+}\pi$ systems where the $\pi$ contribution is offset by $\delta_\pi < 0$.

## 5.2 Comparison with Source Function Decomposition

Bader and Gatti's source function (SF) framework [Bader and Gatti, 1998] provides a different orbital-resolved view of the BCP density. The SF expresses $\rho(\mathbf{r}_b)$ as a real-space integral

$$\rho(\mathbf{r}_b) \;=\; -\frac{1}{4\pi}\int \frac{\nabla^2\rho(\mathbf{r})}{|\mathbf{r}_b-\mathbf{r}|}\,\mathrm{d}^3r, \tag{9}$$

which can be partitioned into per-atom or per-orbital contributions. Farrugia and Macchi [Farrugia and Macchi, 2009] demonstrated orbital decomposition of the SF for transition-metal complexes; subsequent work [Monza et al., 2011, Baggioli, 2018] has extended this to other contexts.

Our approach differs in two key respects: (i) we operate on the local Hessian at the BCP rather than the global SF integral, providing direct interpretation in terms of curvature contributions; (ii) the quantity decomposed—$\varepsilon$ via $\delta_\sigma$ and $\delta_\pi$—is the geometric anisotropy of the density, not the density itself. Both approaches are complementary; the SF probes *which atoms*

*or orbitals contribute electron density to the BCP region*, while our decomposition probes *which orbitals contribute to the two perpendicular curvatures* and hence to $\varepsilon$.

### 5.3 Physical Interpretation: $\varepsilon$ as a $\sigma$-Anisotropy Probe

The systematic pattern $\delta_\sigma > 0$, $\delta_\pi < 0$ (when $\varepsilon > 0$) admits a transparent physical interpretation. The $\sigma$ orbitals at the BCP have non-zero amplitude ($\psi_\sigma(\mathbf{r}_b) \neq 0$) and their second derivatives $\partial^2\psi_\sigma/\partial\xi_q^2$ contribute to the Hessian primarily through the $\psi_\sigma\partial^2\psi_\sigma/\partial\xi_q^2$ term. For an in-plane $\sigma$ orbital with reduced extension along $\mathbf{e}_1$ (out of plane), this term gives $c_\sigma^{(1)} > c_\sigma^{(2)}$, hence $c_\sigma^{(2)} - c_\sigma^{(1)} < 0$ may be expected. However, empirically we find the opposite: $\delta_\sigma > 0$. The mechanism is that in-plane $\sigma$ bonding orbitals (such as C–H bonds) extend significantly along $\mathbf{e}_2$ (in-plane perpendicular) producing a large $|c_\sigma^{(2)}|$, while contributing relatively less along $\mathbf{e}_1$ (out-of-plane). The dominant orbitals contributing to $\delta_\sigma$ are thus the in-plane $\sigma$ frame orbitals, not the C–C $\sigma$ orbital itself.

The $\pi$ orbitals, in contrast, have $\psi_\pi(\mathbf{r}_b) \approx 0$ (the BCP lies in the orbital nodal plane), so the contribution reduces to $2n_\pi(\partial\psi_\pi/\partial\xi_q)^2$. Since $|\partial\psi_\pi/\partial\xi_1| \gg |\partial\psi_\pi/\partial\xi_2|$ (the $\pi$ orbital changes rapidly out of plane and slowly in plane), $c_\pi^{(1)} \gg c_\pi^{(2)} \approx 0$, hence $c_\pi^{(2)} - c_\pi^{(1)} < 0$. This is the origin of the universal $\delta_\pi < 0$ pattern.

The chemist's intuition that "$\pi$ creates ellipticity" captures the kinematic fact that $\pi$ orbitals contribute differentially to $\lambda_1$ and $\lambda_2$ (yielding non-zero $\delta_\pi$), but incorrectly assigns the sign. The correct statement is: the $\sigma$ framework generates a positive $\delta_{\text{tot}}$, and $\pi$ orbitals reduce it.

### 5.4 Limitations

We emphasize several limitations of the present analysis.

**Single-determinant assumption.** The decomposition (Eq. 2) assumes a single Slater determinant or its natural-orbital generalization. For multireference systems (transition states, biradicals, heavy-element complexes with strong static correlation), the classification of orbitals as $\sigma$ or $\pi$ may break down or require generalization to natural orbitals from a correlated wavefunction. Our CCSD/cc-pVDZ results (Table 3) suggest the sign rule is robust to dynamic correlation; static correlation effects remain to be explored.

**Heuristic $\sigma/\pi$ classification.** The geometric criterion (Section 2, $|\psi| < \tau$ + gradient asymmetry) is a heuristic, not a topologically unique definition. The MO-localization independence test (Section 4.3) demonstrates that the qualitative conclusion is robust to MO basis choice. However, quantitative values of $\delta_\sigma$ and $\delta_\pi$ should not be overinterpreted as physical observables independent of the classification choice.

**Degenerate systems.** For cylindrically symmetric bonds ($\varepsilon = 0$), individual $\delta_\sigma, \delta_\pi$ values depend on an arbitrary rotation in the degenerate eigenspace and are not physical. Only the cancellation $\delta_\sigma + \delta_\pi = 0$ is invariant. Our analysis is most informative for non-degenerate $\varepsilon > 0$ systems, the regime where $\varepsilon$ is conventionally used as a chemical descriptor.

**Open-shell systems.** The present work is restricted to closed-shell systems. Extension to ROHF/UHF and to open-shell DFT would require generalization to spin-resolved natural orbitals and is left to future work.

**Heavy-element relativistic effects.** The benchmark omits compounds involving heavy elements where relativistic effects (spin-orbit coupling, scalar relativistic contractions) substantially alter the BCP topology. Application of the framework with relativistic methods (e.g., zeroth-order regular approximation, exact two-component) is straightforward but is not explored here.

**Basis-set range.** The methodological robustness checks span 6-31G* through cc-pVTZ, which covers the practical range for organic chemistry but is not the highest-quality available. The systematic shifts of $\varepsilon$ values between 6-31G* and cc-pVTZ are 5–15%; we expect the sign rule itself to remain unchanged at higher levels but did not explicitly verify with quadruple-zeta or augmented basis sets.

**Sample composition.** The 102-compound benchmark is drawn primarily from organic chemistry (C, H, N, O, F, S, B, P), with limited representation of transition metals, lanthanides, or actinides. Generalization of the sign rule to coordination chemistry, organometallics, and heavy-element systems is an open question.

## 6 Conclusions

We have presented a systematic per-molecular-orbital decomposition of the QTAIM bond critical point density Hessian. The key findings are:

1. *Mathematical identity.* The Hessian admits an exact additive decomposition into per-MO contributions (Eq. 3), with verified zero reconstruction error across all 102 compounds and all methods tested.

2. *Sign pattern.* In 71 of 72 non-degenerate ($\varepsilon > 0$) benchmark cases (98.6%), the $\sigma$ contribution is positive ($\delta_\sigma > 0$) and the $\pi$ contribution is negative ($\delta_\pi < 0$). The single exception is a numerical artifact from BCP localization in a strongly bent bicyclic system.

3. *Methodological robustness.* The sign pattern is preserved across 4 basis sets, 3 theoretical methods, 4 MO localization schemes, and geometry optimization. Numerical magnitudes of $\delta_\sigma$ and $\delta_\pi$ vary by 5–15% across these dimensions, but signs are invariant.

4. *Quantitative confirmation of qualitative pictures.* The result confirms and quantifies Walsh [Walsh, 1949], Cremer [Cremer et al., 1988], and Brown–Bader [Brown et al., 2009] ideas about the role of $\sigma$ orbitals in producing $\varepsilon$ in systems lacking conventional $\pi$ bonds (cyclopropane, $N_2F_4$, $BH_3$, $CHF_3$).

5. *Physical interpretation.* The $\varepsilon$ descriptor in non-degenerate systems is most cleanly interpreted as a probe of $\sigma$-framework geometric anisotropy. The $\pi$ contribution *reduces* $\varepsilon$ rather than producing it.

These findings refine the standard interpretation of QTAIM bond ellipticity and provide a unified analytical framework for interpreting $\varepsilon$ across diverse chemical-bond classes. The methodology is straightforward to extend to other QTAIM descriptors and to additional classes of compounds; we anticipate applications to coordination chemistry, organometallics, and non-equilibrium structures along reaction paths.

## Code and Data Availability

All Python source code, raw computational logs, geometry data, unit tests, and aggregated benchmark results are publicly available at `https://github.com/JackZH26/BCP-Hessian-MO-Decomposition`

(commit `a4d63a4`, accessed 2026-04-26). A comprehensive data archive with persistent DOI is deposited at Figshare, doi:10.6084/m9.figshare.32088900. The repository contains 19 unit tests verifying mathematical reconstruction and sign-rule properties, executable via `pytest tests/`; continuous-integration testing is performed on Python 3.10–3.12. Computations were performed with PySCF version $\geq 2.0$ [Sun et al., 2018, 2020].

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## Author Contributions

J.Z. designed the study, developed the algorithm, performed all computations, analyzed the results, and wrote the manuscript.

## Conflict of Interest

The author declares no conflict of interest.

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